

Public Bike Sharing System

1. Project Overview and Objective

A Public Bike Sharing System enables users to rent and return bicycles at designated stations across a city. It plays an important role in providing convenient, affordable, and sustainable urban transportation. However, managing bike availability across multiple stations is challenging due to fluctuations in demand, uneven bike distribution, and varying station capacities. These challenges can result in stations becoming either completely full or empty, reducing service reliability and negatively affecting the user experience.

This project focuses on evaluating the operational performance and utilization of a city's public bike-sharing network. Using Power BI for data preprocessing and interactive visualization, the project analyzes bike availability, bike stand capacity, station status, banking support, bonus stations, and usage trends. The dashboard provides actionable insights that help monitor station performance, identify underutilized or overloaded stations, and support data-driven decision-making for efficient bike-sharing operations.

2. Data Sources

- **Source Description and Timeline:** Raw Data.xlsx(2022-2025).
- **Domain:** Transportation Analytics

3. Problem Statement

- Analyze bike and bike stand availability across stations.
- Monitor station capacity and utilization to identify frequently full or empty stations.
- Evaluate the operational status of bike-sharing stations, including open and closed stations.
- Compare station performance across different cities or contracts.
- Analyze banking-enabled and bonus stations.
- Identify hourly and daily availability patterns to support efficient bike redistribution.
- Develop an interactive Power BI dashboard that enables stakeholders to monitor key performance indicators and improve the overall efficiency of the public bike-sharing system.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Number	Numeric (Integer)	Unique identification number of the bike station.

Name	String (Text)	Name of the bike station. Usually contains the station ID and location name.
Address	String (Text)	Physical address where the bike station is located
Latitude	Numeric (Decimal)	Geographic coordinates (Latitude) of the station
Longitude	Numeric (Decimal)	Geographic coordinates (Longitude) of the station
Banking	True/False	Indicates whether the station supports bank card payments
Bonus	True/False	Indicates whether the station is a bonus station (special reward/incentive station)
Status	String (Text)	Current operating status of the station
Contract Name	String (Text)	City or bike-sharing service contract under which the station operates
Bike Stands	Numeric (Integer)	Total number of docking stands available at the station (maximum capacity)
Available Bike Stands	Numeric (Integer)	Number of empty docking stands currently available for returning bikes
Available Bikes	Numeric (Integer)	Number of bikes currently available for rent at the station
Date	Date (Long Date)	Date when the station information was last updated
Time	Time (Long Time)	Time when the station information was last updated
Hour	Numeric (Integer)	Hour when the station information was last updated
Bike Utilization %	Numeric (Decimal)	Percentage of bike stands currently occupied by bike

Empty Stand %	Numeric (Decimal)	Percentage of bike stands that are currently empty and available for returning bikes
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5. Tools & Technologies

- **Power BI:** Data cleaning, transformation, Power Query, Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Power BI / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Cleaned and Trimmed Text, handled missing values, standardized formats, and created calculated fields.

Handled Missing Values:

Replacing empty cells in 'Name' column with 'UNKNOWN' text

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Newly Created Calculated Column:

Hour:

Hour = `HOUR(Feuil1[Time])`

Bike Utilization %:

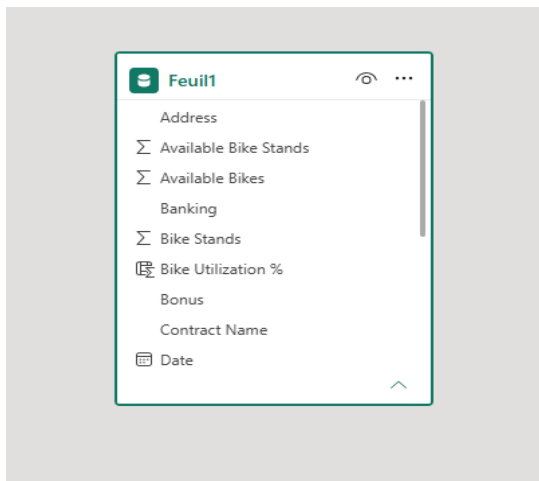
Bike Utilization % = `DIVIDE(Feuil1[Available Bikes],Feuil1[Bike Stands],BLANK())`

Empty Stand %:

Empty Stand % = `DIVIDE(Feuil1[Available Bike Stands],Feuil1[Bike Stands],BLANK())`
`Age","Senior Citizen"))))`

7. Data Modelling and DAX (Power BI)

- **Data Model:** No table relationships were required because the project uses a single table for analysis.



- **Calculated Tables:**

Top 10 Available Bike Stand Station:

Top 10 Available Bike Stand Station = `SELECTCOLUMNS(TOPN(10, Feuil1, Feuil1[Available Bike Stands], DESC), "Station Name", Feuil1[Name], "City Name", Feuil1[Contract Name], "Available Bike Stands", Feuil1[Available Bike Stands])`

Top 10 Available Bike Stations:

Top 10 Available Bike Stations = `SELECTCOLUMNS(TOPN(10, Feuil1, Feuil1[Available Bikes], DESC), "Station Name", Feuil1[Name], "City Name", Feuil1[Contract Name], "Available Bikes", Feuil1[Available Bikes])`

Top 10 Bike Capacity Station:

Top 10 Bike capacity station = `SELECTCOLUMNS(TOPN(10, Feuil1, Feuil1[Bike Stands], DESC), "Station Name", Feuil1[Name], "City Name", Feuil1[Contract Name], "Bike Capacity", Feuil1[Bike Stands])`

- **DAX Measures:**

1. Total Available Bike Stands = `SUM(Feuil1[Available Bike Stands])`
2. Total Available Bikes = `SUM(Feuil1[Available Bikes])`
3. Total Unavailable Bike Stands = `CALCULATE(COUNTROWS(Feuil1), Feuil1[Available Bike Stands]=0)`
4. Total Unavailable Bikes = `CALCULATE(COUNTROWS(Feuil1), Feuil1[Available Bikes]=0)`

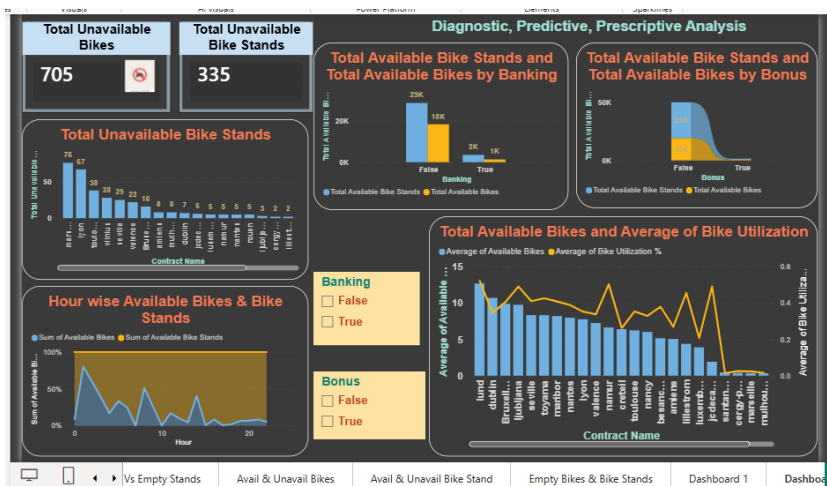
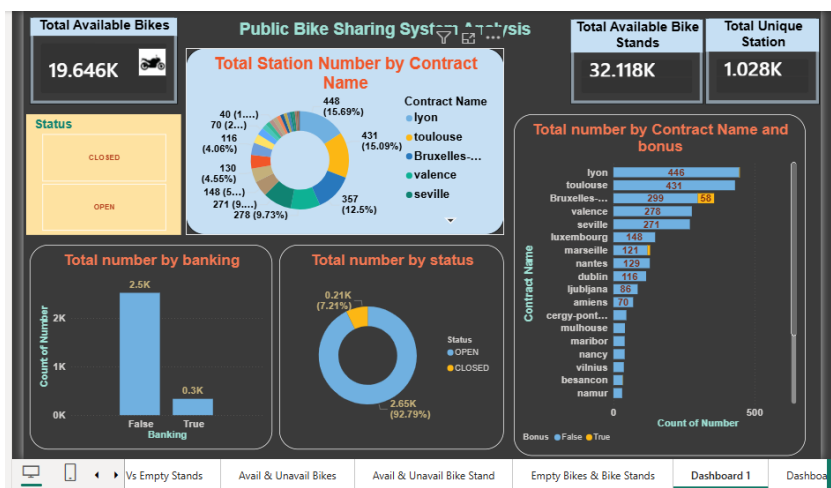
5. Total Unique Station = `DISTINCTCOUNT(Feuil1[Number])`

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Developed multiple visualizations (bar charts, column charts, combo charts, line charts, pie charts, area chart, cards, and tables) based on the problem statement to communicate key insights.
- Designed an interactive Power BI dashboard using drill-downs, filters, slicers, and clear titles to enhance data exploration and user experience.

Dashboard Screenshot:



9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data

- **Identifying key insights from the analysis**

- **Descriptive**

1. The bike-sharing network consists of **2,856 station records** across **1,028 unique stations**, with **57,203 total bike stands**, **19,646 available bikes**, and **32,118 available bike stands**.
2. Approximately **92.79% of stations are operational (OPEN)**, while **7.21% are CLOSED**, indicating high overall service availability.
3. **Lyon** has the largest bike-sharing network with **448 stations**, followed by **Toulouse (431)** and **Bruxelles-Capitale (357)**.
4. Most stations are **non-bonus** and **do not support banking**, showing that advanced station features are available only at a limited number of locations.
5. The highest bike and bike stand availability is observed around **14:00 (2 PM)**, suggesting that most data snapshots were recorded during this period.

- **Diagnosis**

1. Cities with larger station networks, such as **Lyon** and **Bruxelles-Capitale**, naturally report higher numbers of available bikes because they have significantly greater station capacity.
2. **Marseille** records the highest number of unavailable bikes and unavailable bike stands, indicating frequent shortages and demand imbalance or insufficient bike redistribution.
3. In almost every contract, **available bike stands exceed available bikes**, suggesting that many bikes are currently in use or are being redistributed. Until those bikes are returned, the stations will have more empty stands.
4. **Dublin** and **Luxembourg** also have high empty stand percentages, indicating that additional bikes could be supplied if demand supports it.
5. **Namur (41%)** and **Lund (46%)** have comparatively lower empty stand percentages, reflecting better utilization of station capacity.
6. **Bonus stations** perform slightly better than regular stations, averaging **8.53 available bikes** compared with **6.84** at non-bonus stations, suggesting bonus incentives may help maintain better availability.
7. Stations with **banking facilities** have different availability patterns, but because banking stations are relatively few, the overall impact is limited.
8. **Dublin** combines high average bike availability with strong utilization, making it one of the most efficiently performing contracts.

- **Predictive**

1. **Marseille** is likely to require the earliest bike redistribution because it has the lowest average bike availability and the highest number of unavailable bikes.
2. High-capacity cities such as **Lyon**, **Toulouse**, and **Bruxelles-Capitale** are expected to continue experiencing high demand because they contain the largest station networks and bike capacities.

3. Stations with consistently low utilization, such as **Vilnius, Rouen, Mulhouse, and Cergy-Pontoise**, are likely to remain underutilized unless demand patterns change.
4. **Lund** has the highest bike utilization (52%), followed by **Namur (50%), Ljubljana (49%),** and **Lillestrom (46%)**, indicating strong demand relative to capacity and are likely to become empty during peak demand periods unless bikes are redistributed proactively.
5. Cergy-Pontoise (92%), Santander (84%), and Mulhouse (76%) have very high empty stand percentages, indicating that most docking spaces are vacant because relatively few bikes are available at these stations. This may suggest low bike availability, an imbalance in bike distribution, or underutilization of the station's capacity.
6. **Dublin** combines high average bike availability with strong utilization, making it one of the most efficiently performing contracts
7. Stations showing recurring shortages of available bikes or docking spaces are likely to require more frequent balancing operations.
8. If current trends continue, non-banking and non-bonus stations may continue to offer a lower level of service compared to stations equipped with these features.

- **Prescriptive**

1. Prioritize **bike redistribution** to stations and cities with consistently low bike availability, especially **Marseille** and other high-demand locations.
 2. Expand **bike stands or add new stations** in high-capacity cities such as **Lyon, Toulouse, and Bruxelles-Capitale** to accommodate increasing demand.
 3. Investigate low-utilization stations such as **Vilnius, Rouen, Mulhouse, and Cergy-Pontoise** to determine whether bike redistribution, relocation, or station resizing is needed.
 4. Increase the number of **bonus stations** in busy areas, as they show better average bike availability than non-bonus stations.
 5. Use hourly availability trends to schedule bike redistribution before peak demand periods, ensuring bikes and empty stands are available when users need them most.
 6. Expand banking facilities at strategically important stations if operational data continues to show better performance at these locations.
 7. Implement **dynamic bike redistribution** to move bikes from low-demand stations to high-demand stations before shortages occur.
 8. Monitor stations with high bike utilization and high empty stand percentages in real time to prevent stations from becoming completely empty or full.
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10. Conclusions

The Public Bike Sharing System Analysis demonstrates how Power BI can transform operational data into actionable insights. The analysis identified high-demand cities, station utilization patterns, bike availability, and operational challenges such as bike shortages and underutilized stations. These insights can help optimize bike redistribution, improve station management, and enhance the overall efficiency and reliability of the bike-sharing network.